

A detailed black and white illustration of a bamboo stalk with several leaves, positioned vertically on the left side of the slide.

CRS Approach on Bamboo

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Humanitarian bamboo



Bamboo
training
Manual

Pilot
workshop in
the
community

**Bamboo for
reconstruction**

Effective
treatment
machine

Construction
techniques

Bamboo
pest survey
And study

Design of
bamboo in
the Demo
house

Survey in areas
of Gorkha

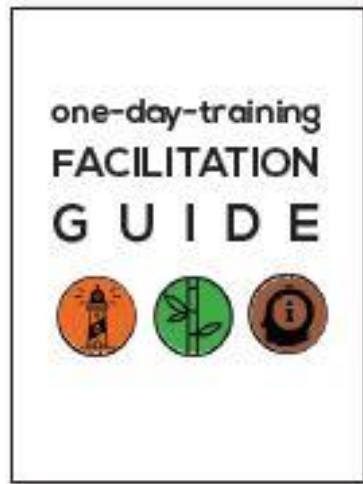
Construction and
design

Development of
key message

Treatment
machine

PESTS

Bamboo training manual and Guides

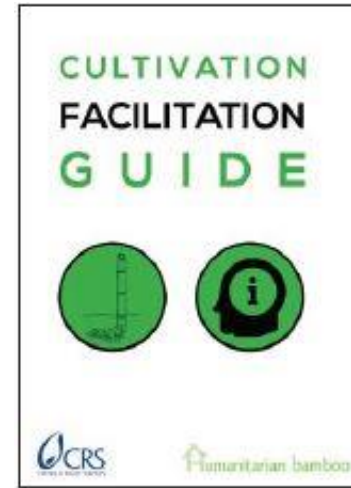
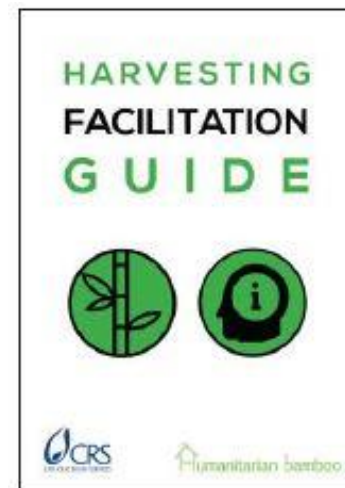


Bamboo and Cultivation

- Basic on bamboo methods of cultivation

Harvesting

- Harvesting methods WHEN and HOW

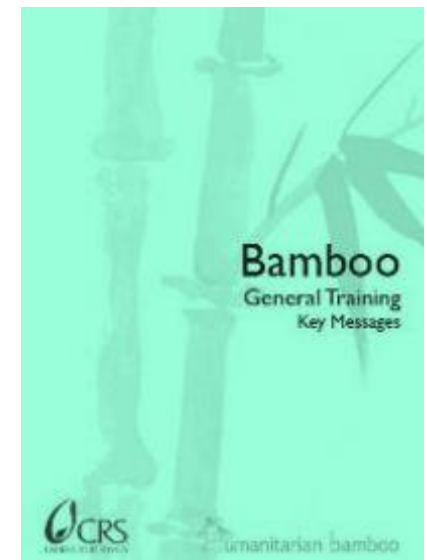


Treatment

- Treatment methods WHY and HOW

Construction and Design

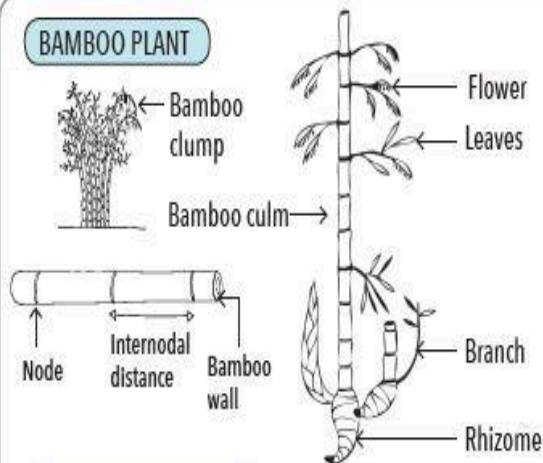
- Techniques on connection and design perspective



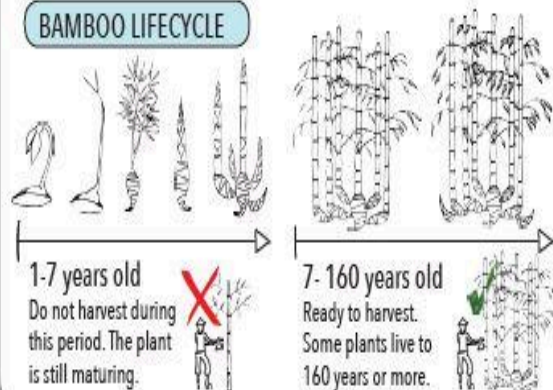
BAMBOO AND CULTIVATION

Care for your bamboo. A healthy plant will produce high quality bamboo that can be regularly harvested. A sick plant will produce low quality bamboo and can encourage pests.

BAMBOO PLANT



BAMBOO LIFECYCLE



WHEN TO PLANT

For best results, plant at the start of the wet season before heavy rains.



Dry season



Start of the Wet season



Heavy rain has started

CULTIVATION METHODS

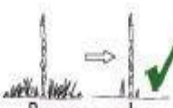
Plant care
Regularly check and care for new plants



Protect from animals

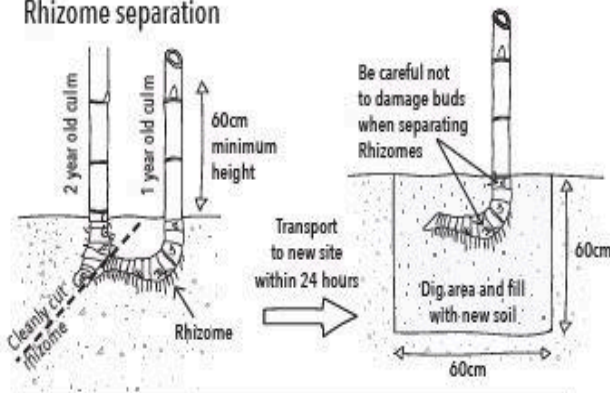


Water well after planting



Remove nearby weeds

Rhizome separation



Cuttings

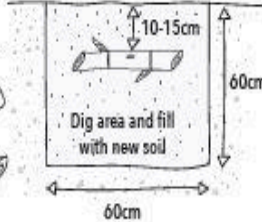
Take cuttings from 2-3 year old culm with fresh buds or shoots.

Make a clean cut, 5cm from the node



Branch shoot

Cut hole and fill with water



HARVESTING BAMBOO

Harvest your plant after the wet season has finished. Know how and when to harvest bamboo culms to ensure your plant remains healthy and productive.

WHEN TO HARVEST

Harvest early morning in the dry season



Wet season

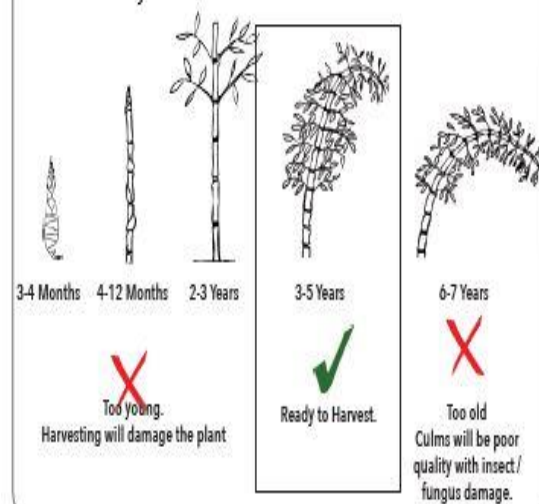


Dry season



Morning

Harvest 3-5 year old culms.



HOW TO HARVEST

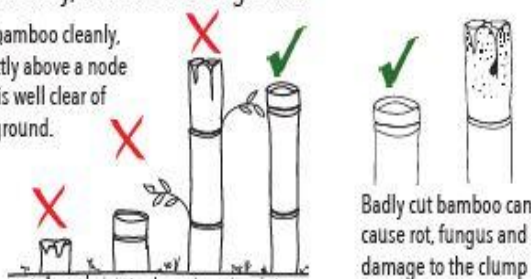
Only ever harvest 1/3 of a clump



Don't harvest more than one third of your bamboo clump at once.

Cut cleanly, well above the ground

Cut bamboo cleanly, directly above a node that is well clear of the ground.



Badly cut bamboo can cause rot, fungus and damage to the clump

Remove old, damaged and infected culms

Shoot Borer moths create recognisable holes above the node. Bamboo culms should be removed to reduce infestation of the clump. Old bamboo (7 years and older) should be removed, along with broken or damaged culms.



TREATING BAMBOO

If treated and handled correctly, bamboo can last more than 20 years, but only if your structure is designed well. It can be used as an alternative to timber in permanent housing.

WHY TREAT?



Treatment stops pests from destroying bamboo. It can increase the lifespan of bamboo dramatically.



Treated bamboo in permanent construction

Well designed bamboo constructions roofing, flooring and walling can last more than 20 years.



Untreated bamboo in temporary structures

Your structure will only last maximum 1-5 years if you use untreated bamboo



HOW TO TREAT BAMBOO

There are many common treatment methods available, some include:

Boucherie Treatment

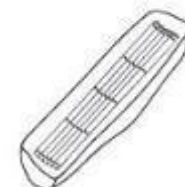
- Portable and can run off power, petrol or hand pump
- Treat up to 8 bamboo culms at once
- You must treat bamboo within 24 hours of harvesting
- Takes 1-2 hours (depends on species)



Soaking Pools

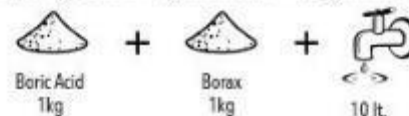
Holes must be drilled between each node OR node membranes broken with a long pole.

- Can treat many bamboo at once
- Treat soon after harvesting if possible
- 3 Weeks to treat
- Pools can be used to treat timber



TREATMENT MIXTURE

Mix borax powder and boric acid powder with water in a ratio of 1:1:10. (For example: 1kg of Borax + 1kg of Boric Acid + 10 kg/liters of water).



DESIGN

Careful design of a structure is very important in bamboo construction. A poorly designed structure can have a dramatically reduced lifespan, while a well designed bamboo structure can last over 20 years.

CHOOSING YOUR BAMBOO

You can use many different species of bamboo for one structure.

Bamboo species

Thin walled bamboo: suitable for split bamboo, rafters, purlins.



Thick walled bamboo: suitable for post and structure



Solid bamboo and Nigalo: suitable for purlins, handicrafts and tool handles.



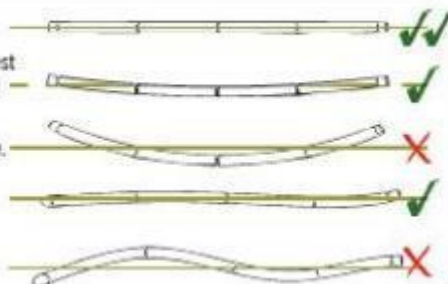
Splits

Make sure splits don't go through the nodes. If you have to use cracked bamboo, tightly bind the node with lashing or clamps.



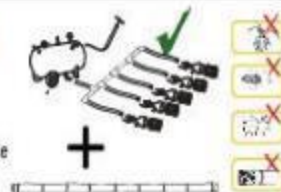
Straightness

Use only the straightest bamboo for structural purposes.
Use string to measure.



USE TREATED BAMBOO

Use treated bamboo for permanent construction. It will protect against pests and rot and dramatically increase the lifespan of your structure.



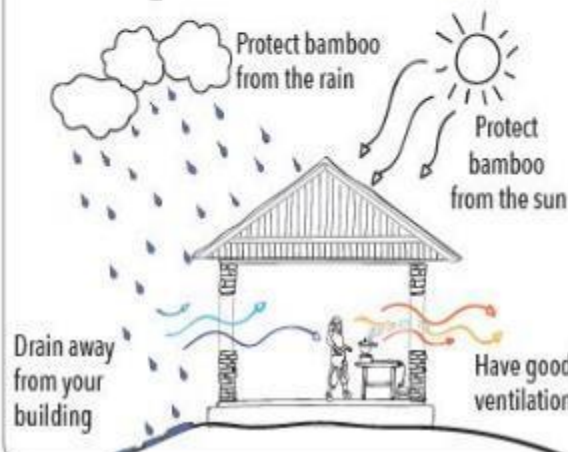
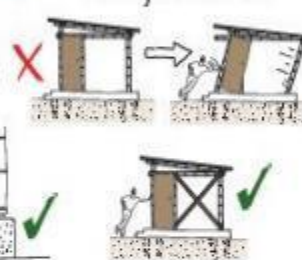
LONG LASTING DESIGNS

To ensure the longevity of your structure, make sure bamboo is protected from the sun, rain and ground.

Raise off the ground



Brace your structure



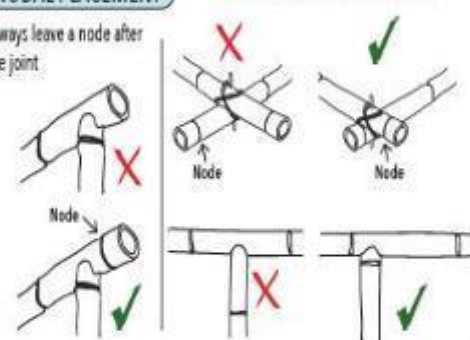
CONSTRUCTION

If treated and handled correctly, bamboo can be used as an alternative to timber in permanent housing. It can last more than 20 years.

NODAL PLACEMENT

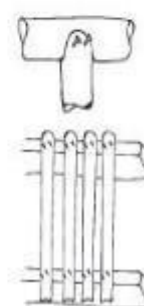
Join bamboo close to the nodes.

Always leave a node after the joint



NAILS

Low difficulty
Fast
Splits easily
Low strength



Splitting

Nails can easily split your bamboo

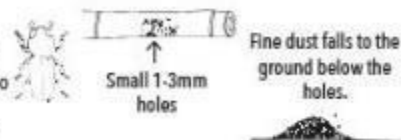


PESTS AND INDICATORS

** This is a rough guide only and insects will vary depending on area

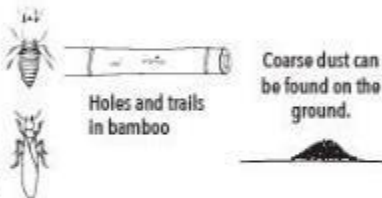
Powder post borer

The pinhole borer creates small holes in the bamboo skin while eating the inner flesh and gradually weakening its structure.



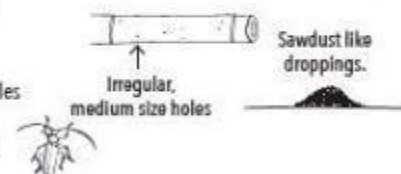
Termites

Termites aren't always obvious until major damage has been done to the internal flesh of the bamboo. Once this damage is done, the culm weakens and loses all structural strength.



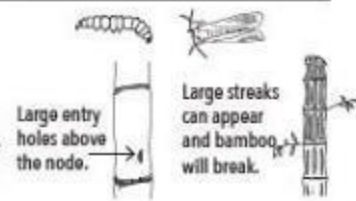
Long horned beetle

Can tunnel through the entire bamboo culm, leaving medium sized holes throughout. Can cause major structural damage.



Shoot borer moth

Attacks living bamboo clumps. Enters through young shoots. Leaves a recognisable hole just above the node. Infected culms should be removed from the clump.



INSPECTING FOR PESTS AND DAMAGE

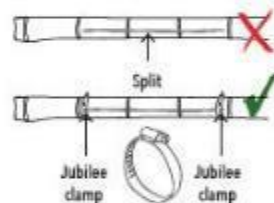
Even when treated, your bamboo structure could still be at risk of pests. Inspect your structure regularly, and replace infected bamboo or treat if possible.

REGULARLY CHECK FOR DAMAGE

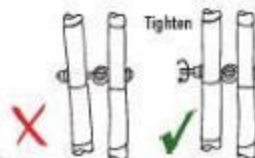
Inspect for infected culms. Regularly inspect your building for pests. Damaged bamboo may indicate your structure isn't using treated bamboo. Replace infected culms with new treated culms.



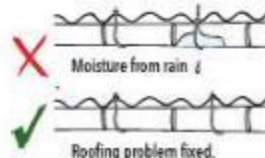
Check for cracks and damage. Jubilee clamps can be used on minor cracks. If badly damaged, replace bamboo to avoid further damage.



Tighten your connections. Lashing and bolts can loosen over time. Check your connections and tighten or replace parts if needed.



Check for wet bamboo. Bamboo shouldn't be exposed to rain or moisture. It will quickly rot and degrade, even when treated. Check for leaks, rot and moisture in bamboo culms.



LASHING

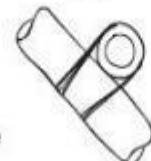
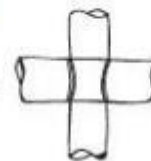
Use jute lashings, nylon string or wire

Cheap

Flexible connection

Time consuming

Can loosen over time



LASHING AND PEGS

High strength

Time consuming

Can loosen over time



Regularly check your lashing. Lashing can loosen over time and bamboo can shrink. Make sure you regularly check and tighten or replace your lashing.



BOLTS

High strength

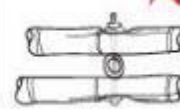
Durable

Can cause crushing

Requires drilling

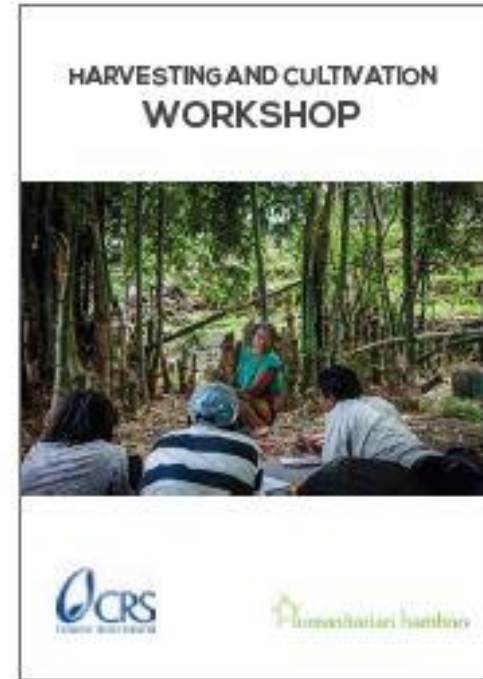
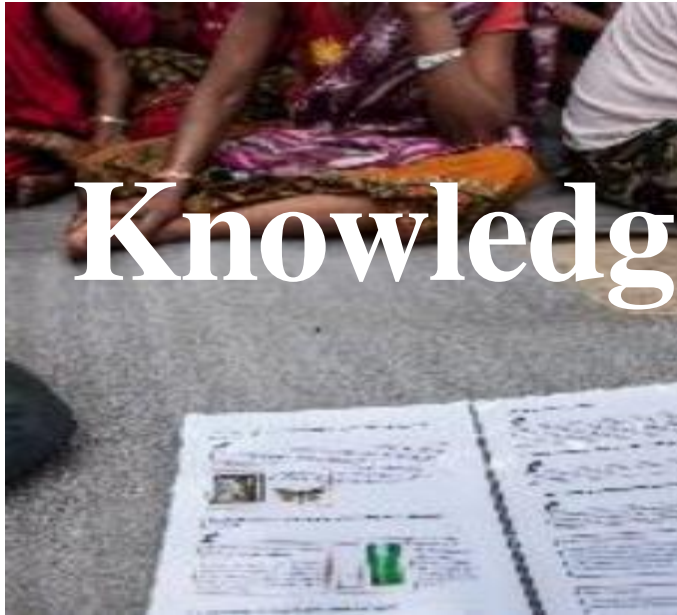


Avoid crushing from over tightening. Don't over tighten the bolts and crush your bamboo



Pilot workshop in the community on the Key messages

Knowledge Transfer



Boucherie treatment machine

Research and advancement



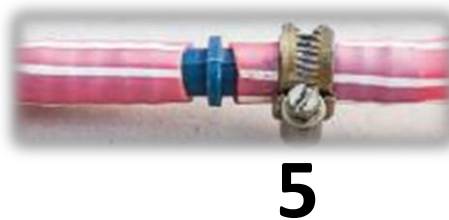
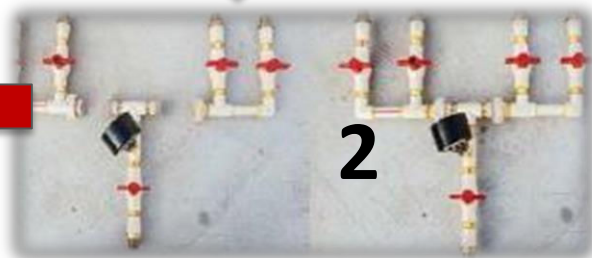
Community

Mechanical machine

- Removal of metal compressor
- Custom built metal elements replaced with common plumbing fittings

Gravity Boucherie system

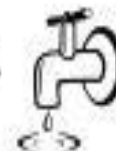
Assembling the parts



± kg.



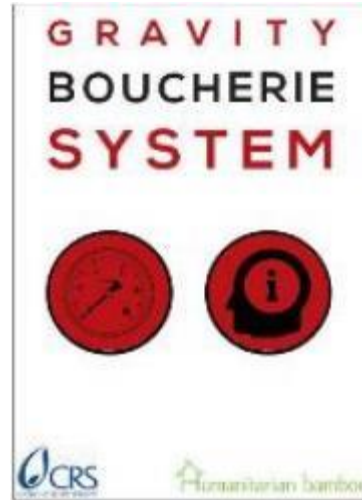
± kg.



20 lt.

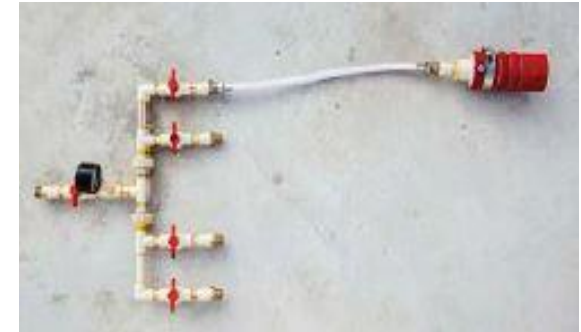


Assembling with the help of catalogue

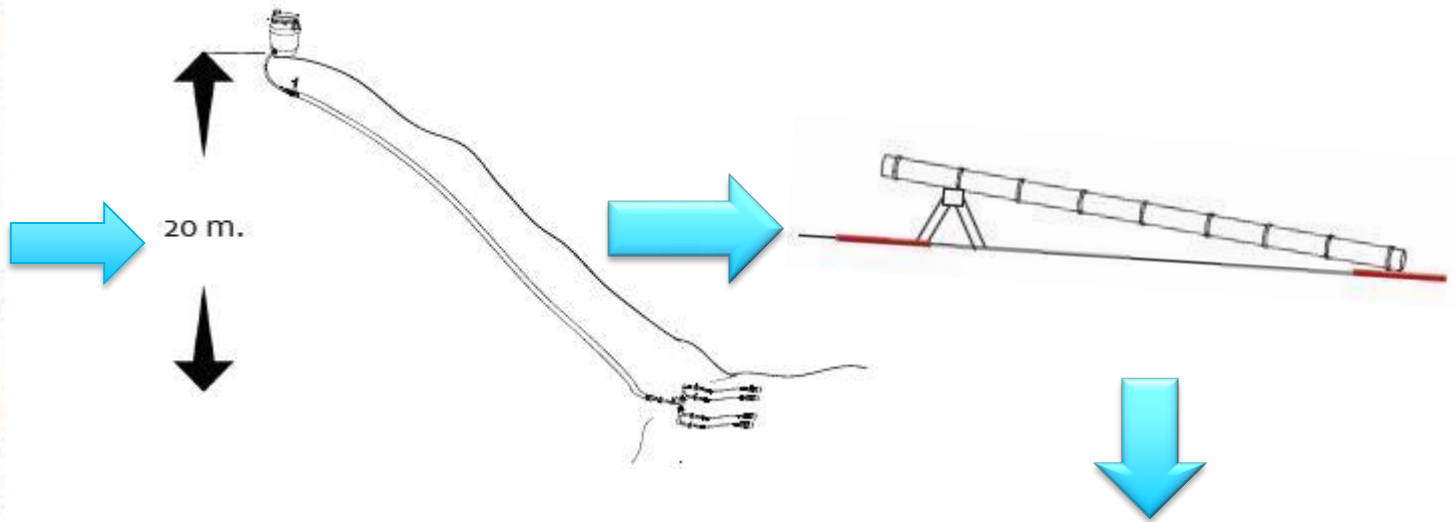


1. BOUCHERIE KIT COMPONENTS.

We've provided a Gravity Boucherie System (GBS) kit. Please check that all the pieces mentioned in this chapter are in your gallery.



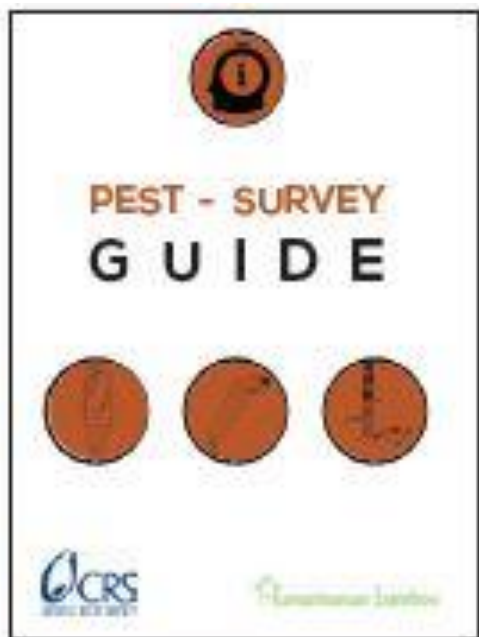
Treatment process



Pests Guide and Workshop



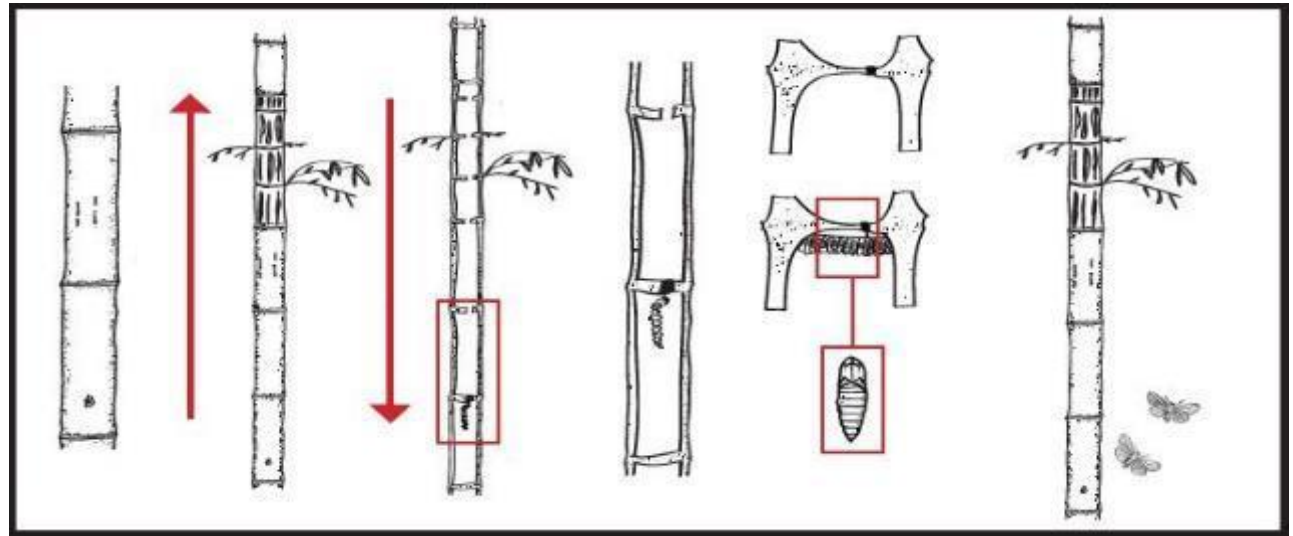
Bungkot VDC, Gorkha



Pests survey



Life cycle of the pest

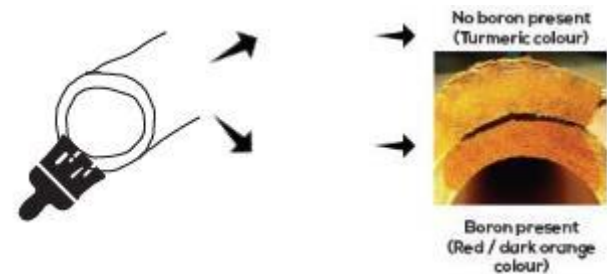
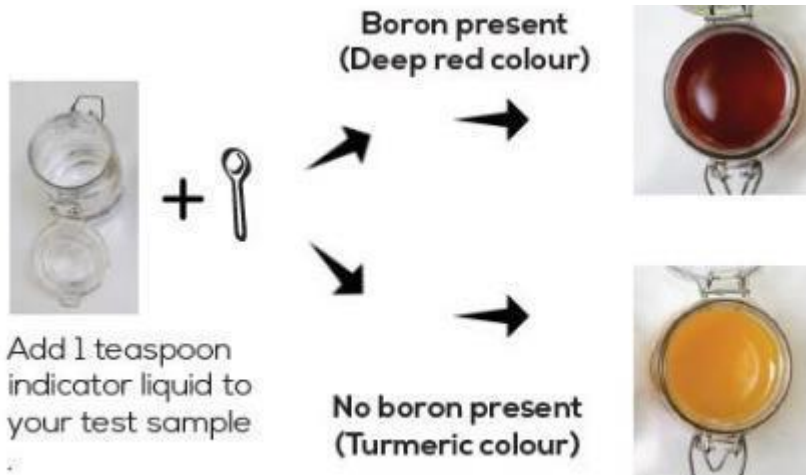


- During the last months, different field activities inside CRS bamboo program revealed the presence of a characteristic insect attack in a high percentage of the bamboo clumps (more than 90%).
- In different wards of Bungkot and Taple, a severe damage was observed in multiple culms of the clumps inspected. After an initial inspection, this insect was identified as a shoot borer moth from the Noctuidae family, the Pareuplexia.

Pests survey



Testing boron in liquids and Bamboo



Challenges

Community
perspective

- *Community behavior and change (tradition belief and local practice)*

Government
guidelines
and codes

- *Calculation and testing to get the approval*
- *No Bamboo house in the design*

Bamboo Demo House



Bamboo Demo House



Basis of Structural Design

Parameter	RCC Structure	Steel Structure	Bamboo Structure
Engineering Knowledge Base	Strong	Strong	Weak
Method of Design	Limit State Method	Limit State Method	Working Stress Method
Self-Weight	High	Low	Low
Critical Dynamic Loads	Seismic	Wind	Wind
Code of Practice	IS-456	IS-800	IS-15912

Comparison of Material Properties

Parameter	Concrete (M20)	Structural Steel (E250A)	Structural Bamboo (<i>B.balcooa</i>)
Ultimate Compressive Strength	20 MPa	410 MPa	46.7 MPa
Ultimate Tensile Stress	~3 MPa	410 MPa	65.4 MPa
Shear Strength	1.2 MPa	144 MPa	8.9 MPa
Modulus of Elasticity	22.36 GPa	200 GPa	7.93 GPa



DUDBC Approval Process

1. Completed Drawings
2. Full Structural Report:
 - Structural Values and Properties used for Design
 - Critical Element Stresses Report
 - Report full hand calculations on joint detailing
 - Citations/References to academic journals

Q & A

Thank you

